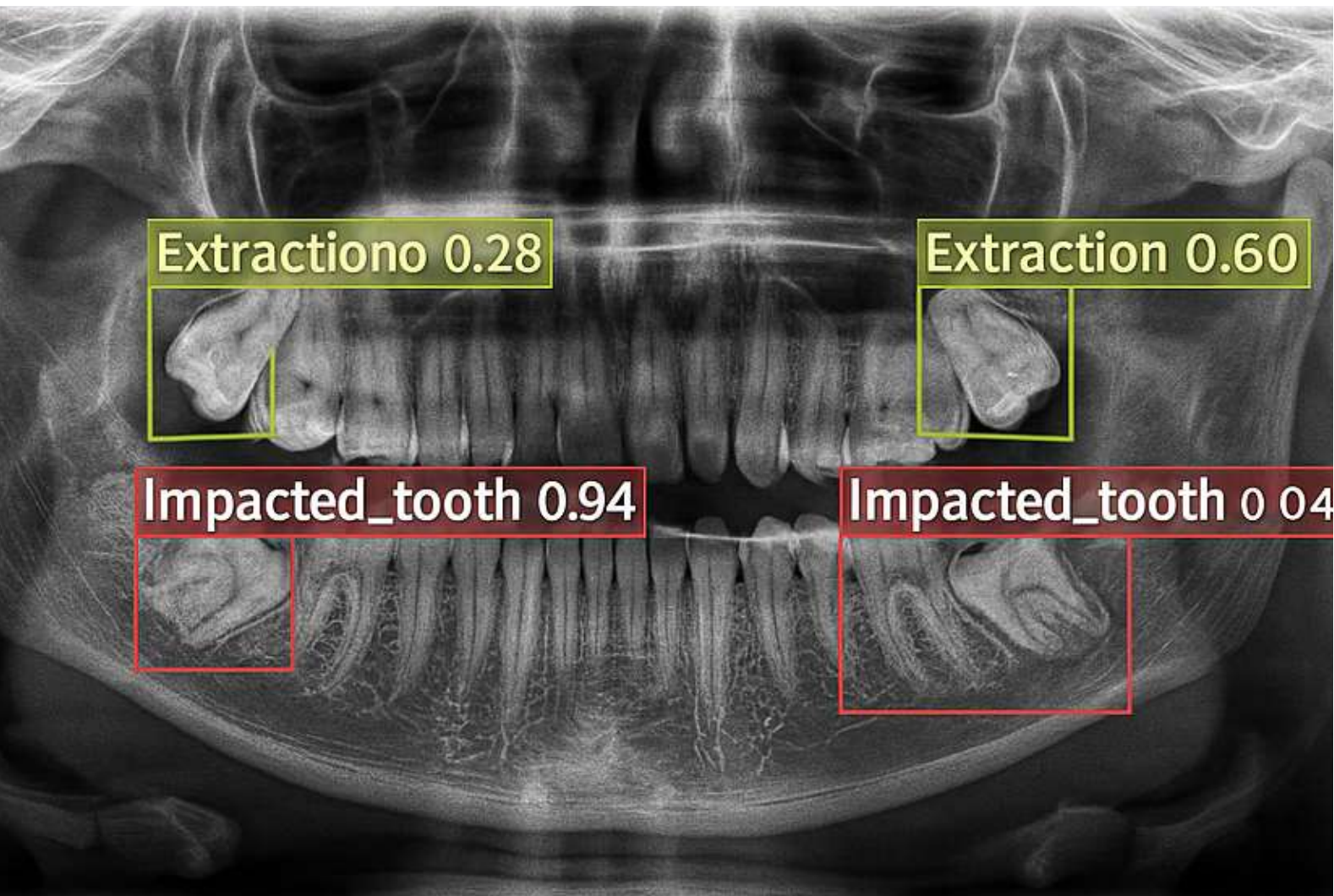


Dental Anomaly Detection through Biomedical X-rays: Deep Learning Approach



Department of Computer Engineering

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Typical types of Dental Anomalies ?

Abnormalities Detection Types:

- Impacted_tooth
- Filling
- Implant
- Root Canal
- Extraction
- Crown



The problem / Current Issues:

Disadvantages of Manual Interpretation:

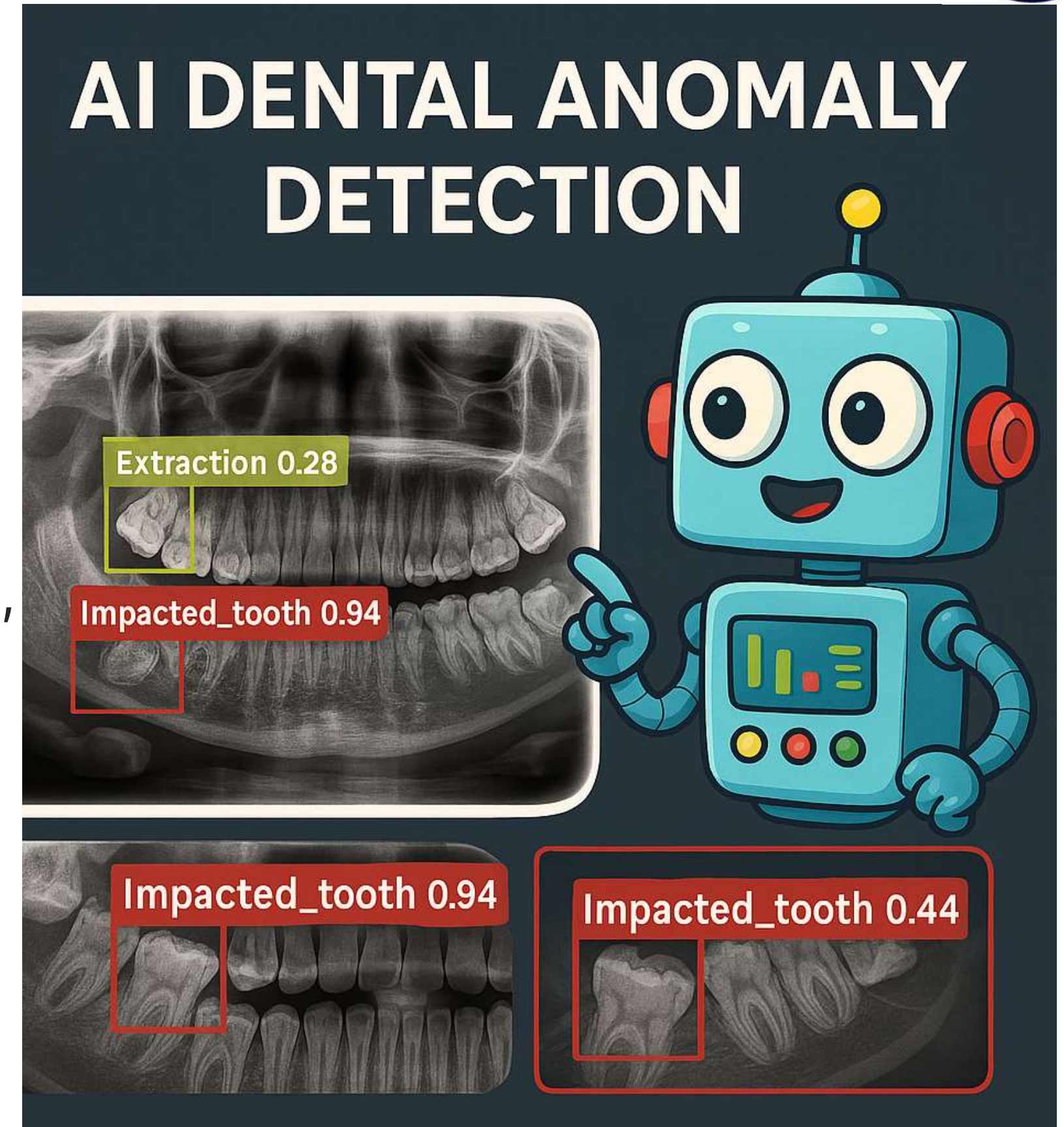
1. Time-Consuming & Inefficient
2. Human Error & Inconsistency
3. Accessibility Issues



Why to Automate the process?

Benefits of Automating this process:

1. Automating detection → Faster, cheaper diagnoses.
2. Assisting dentists → Lets them focus on treatment, not detection.
3. Scalable → Can screen thousands of X-rays/day.
4. More Accurate → 95.3% mAP vs. human error rates.



Note: AI Generated Image (Copilot)

Research Purpose



Develop AI System

Create an accurate AI system for dental anomaly detection.



Reduce Errors

Reduce diagnostic errors.



Faster Screening

Develop a faster, more reliable screening method. Improve early detection of dental pathologies and patient care.

Technology Used

YOLOv8: You Only Look Once version 8

Python: Jupyter Notebook (Image Processing and model training)

Flask: Python framework, enabling integration between YOLOv8 and web interface.

User Interface: HTML and CSS



Flask



python



Methodology: Dataset and YOLOv8

Data Annotation

800 images with annotations

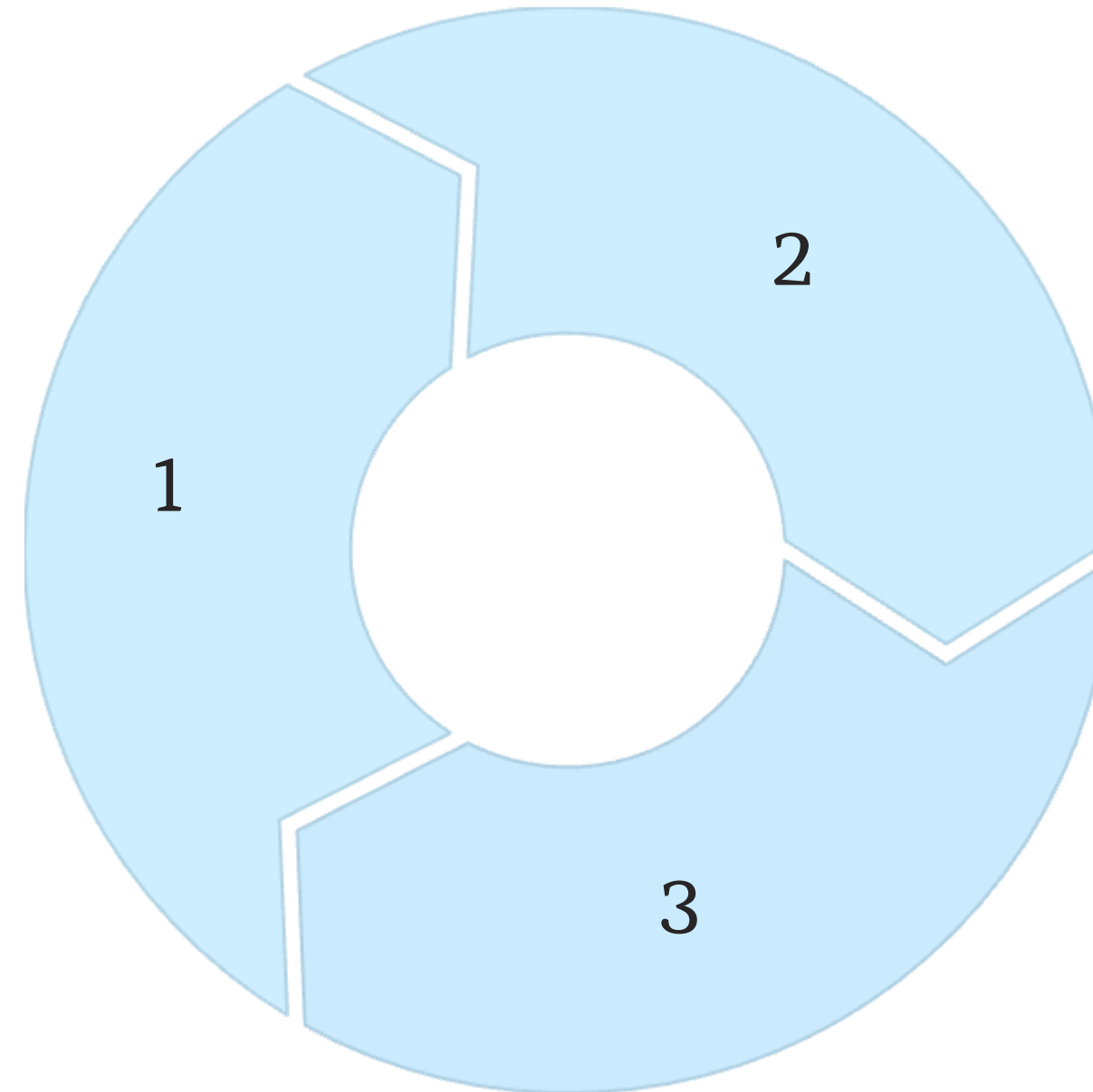


Image Processing & YOLOv8

Utilized for real-time detection.

Deployment

Flask API and User Interface.

YOLOv8 Official Documentation: <https://yolov8.org/yolov8-documentation/>

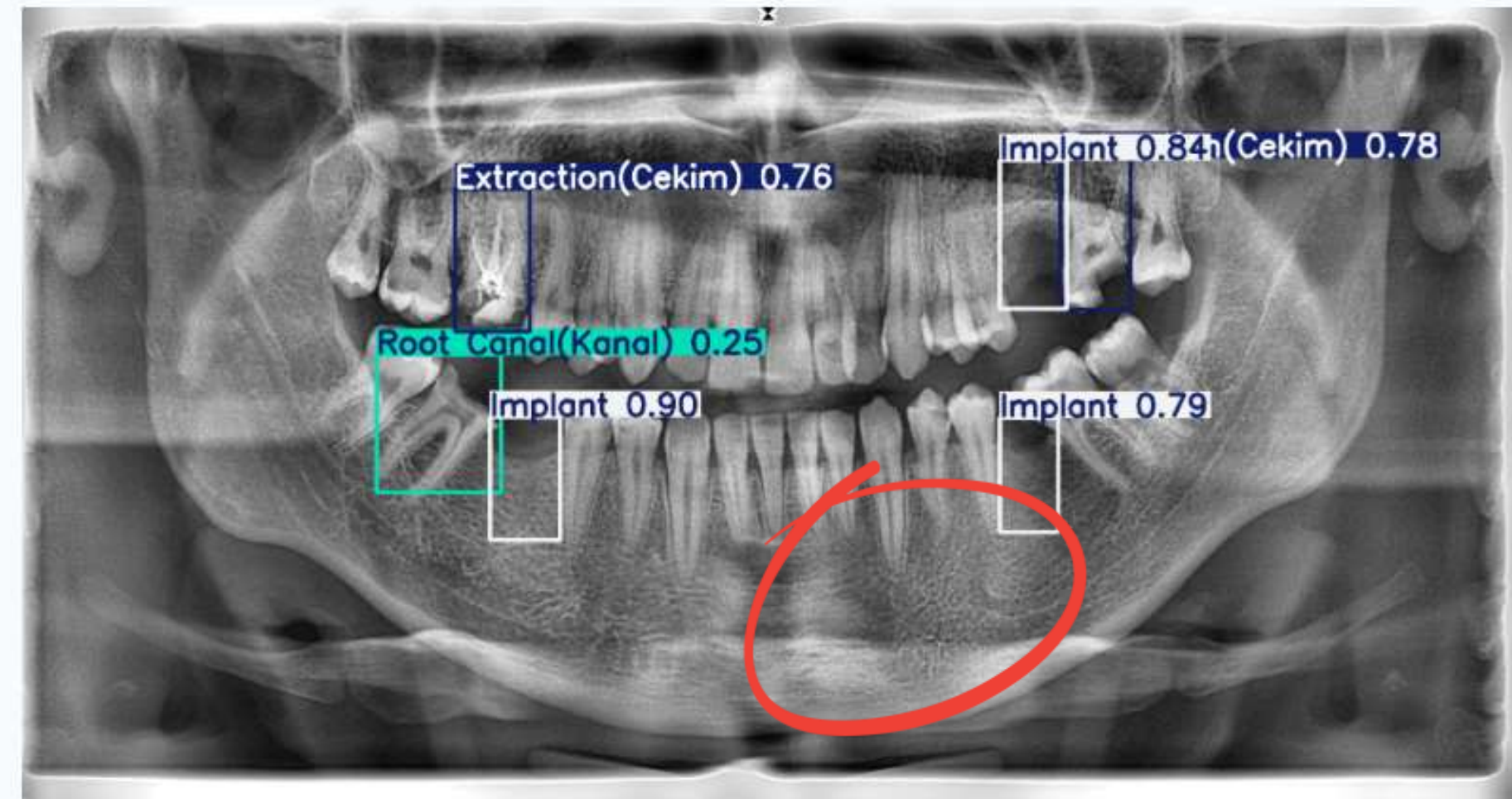
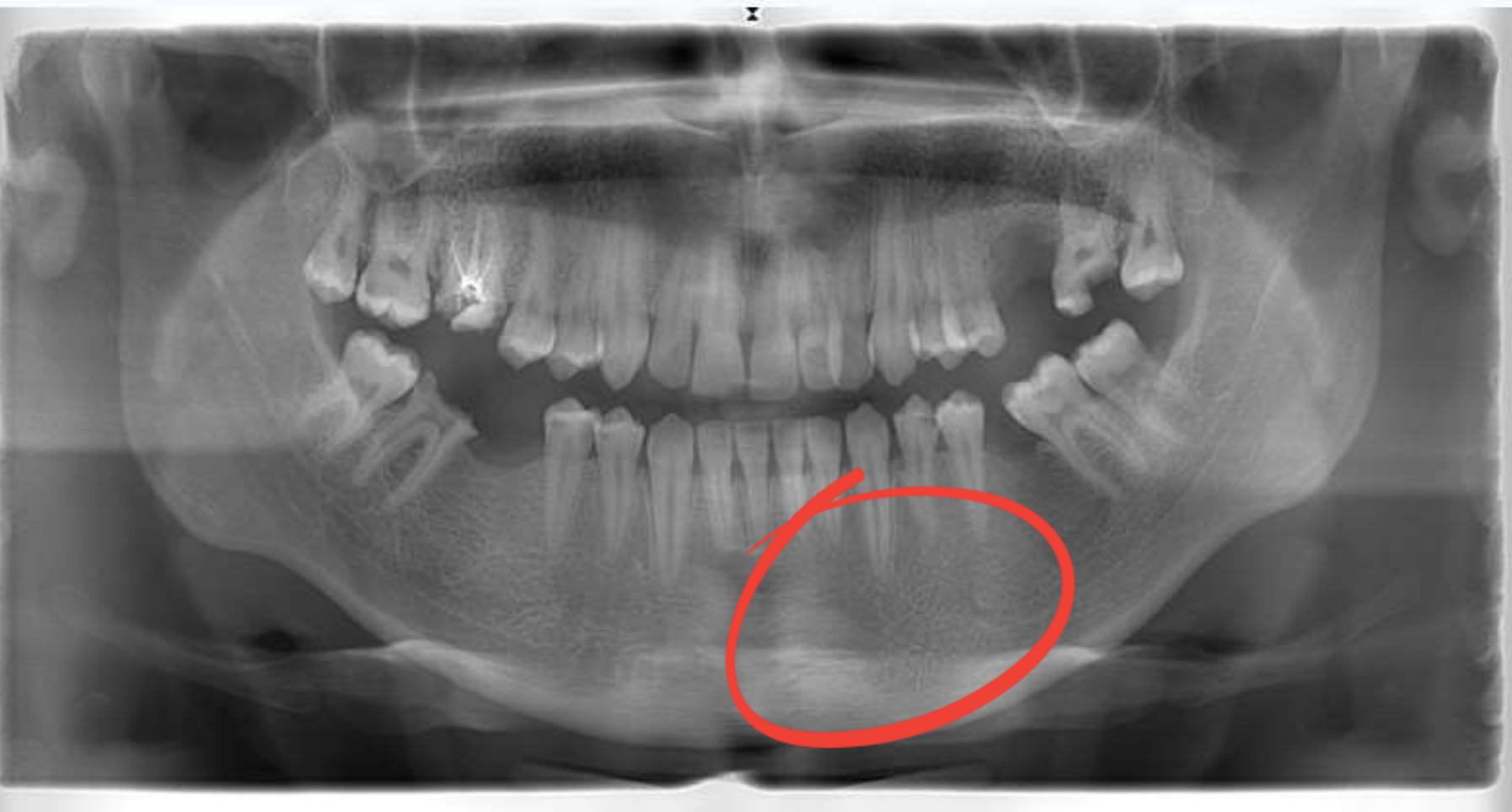
CLACHE (Contrast Limited Adaptive Histogram Equalization)

Before

After

 Original Image

 Detected Anomalies



- Joudet, C. (2018). Deep Learning on Dental X-rays. Medium. [Link](#)

Training the model



Used Local Machine
72 hours of training moreless.



Early Stopping
Stopped after 572 epochs.



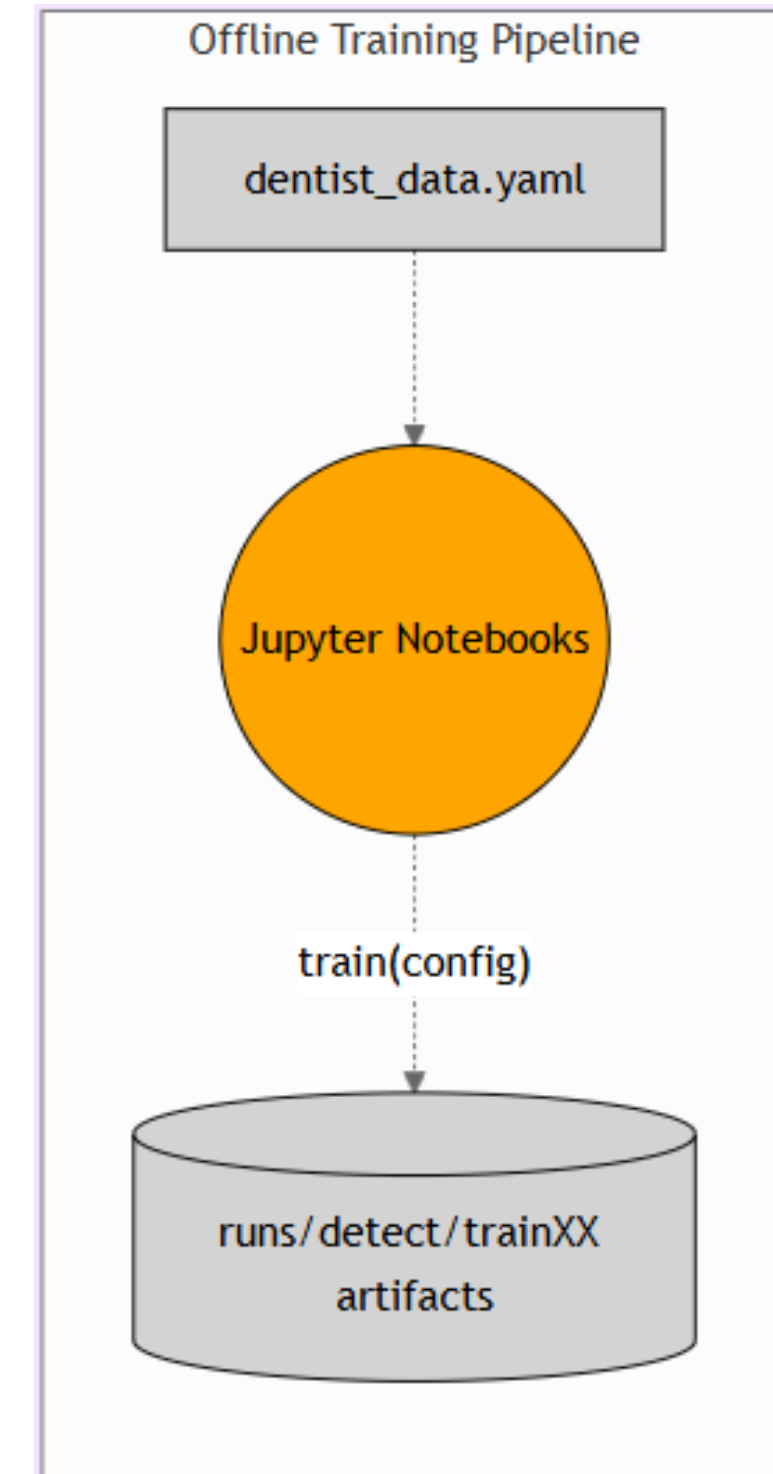
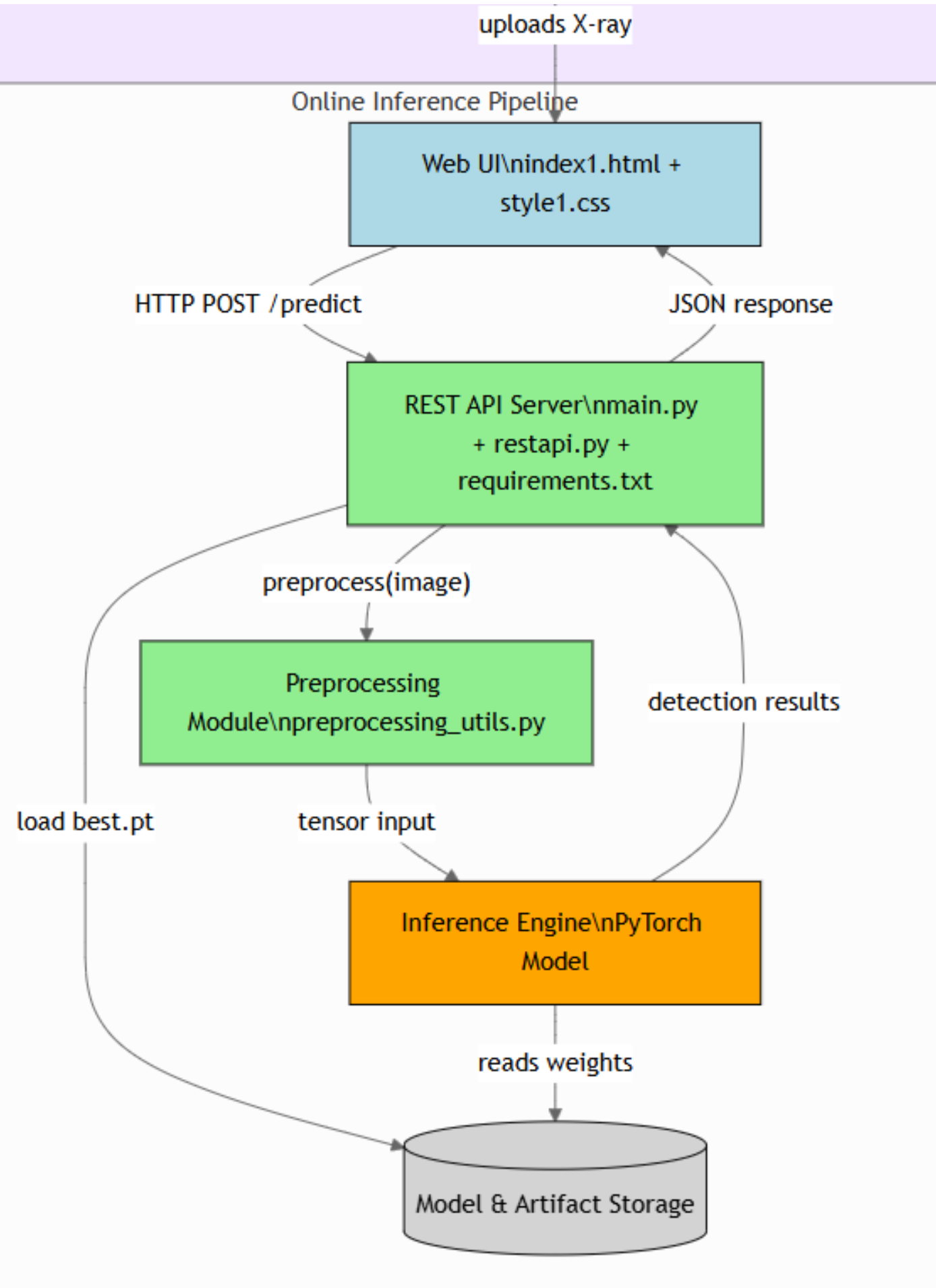
Saving the Best and the Last models
best.pt at the 561 epoch.



Early stopping is a regularization technique used to prevent overfitting by stopping the training process once the model's performance starts to degrade.

Project Architecture

1. Training Pipeline
2. Web User Interface
3. REST API server
4. Image Resizing and CLACHE
5. Read best.pt weights
6. Detect Anlomalies
7. Response to UI through REST API



Performance Metrics

Accuracy Vs. Number of instances:

95.3%

mAP

Mean Average Precision.

96.7%

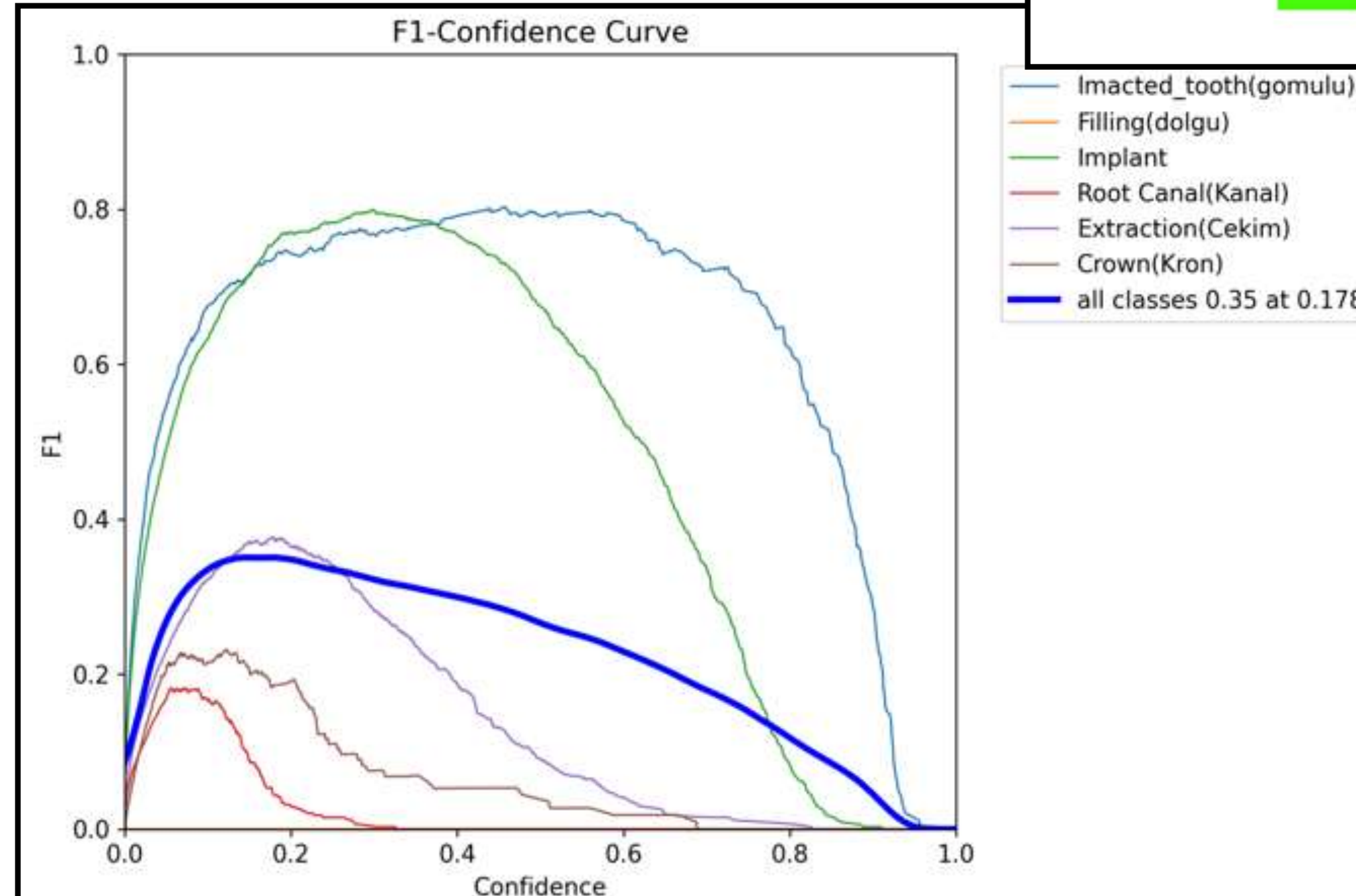
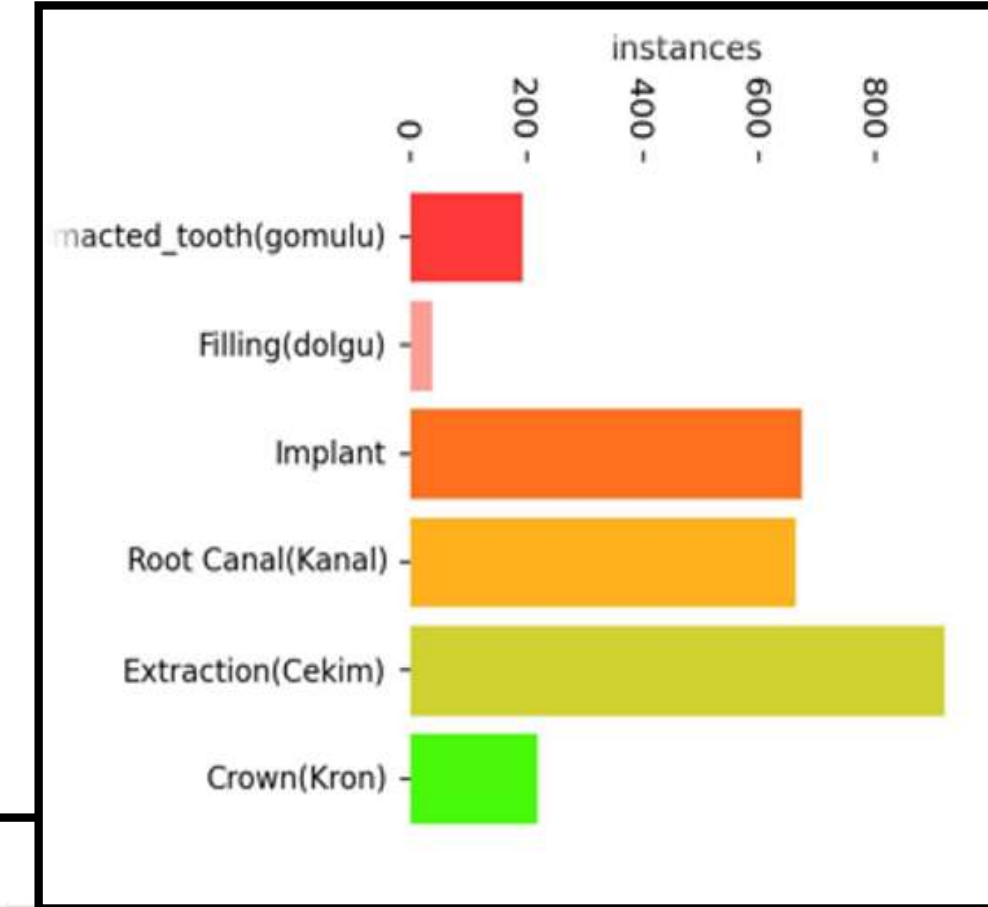
Precision

Accuracy of positive predictions.

93.3%

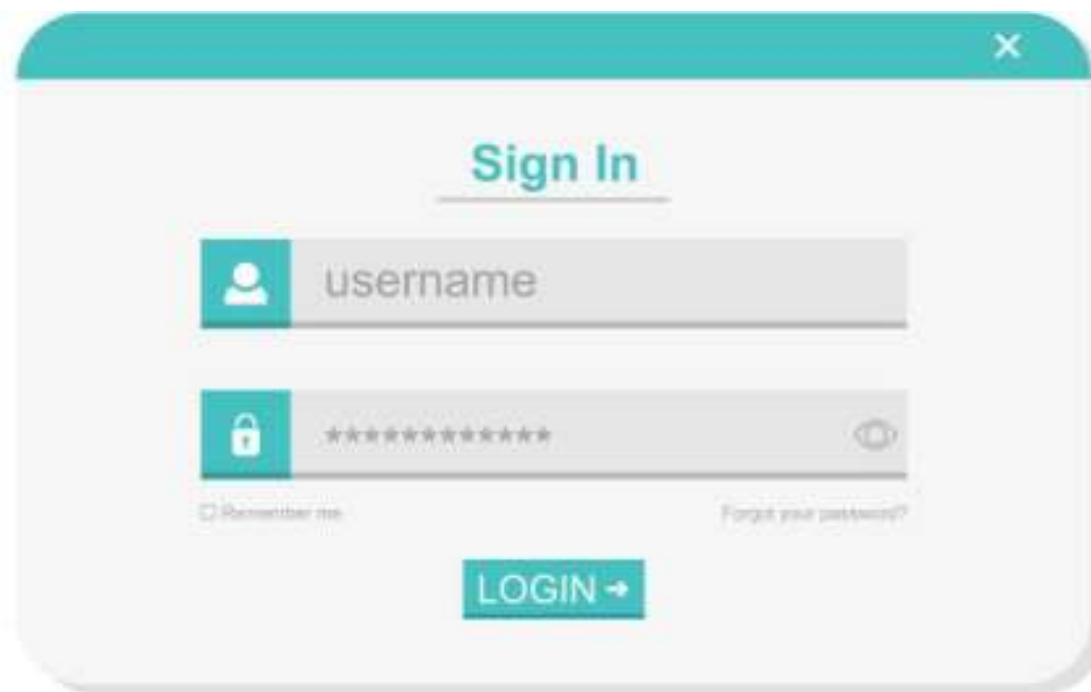
Recall

Ability to find all relevant cases.



What about the Business side?

The System might be sold as monthly/yearly subscription for clinics so the clinic can immediately use the system right after taking the X-ray image from patient.



Sign In

username

password

☐ Remember me

[Forgot your password?](#)

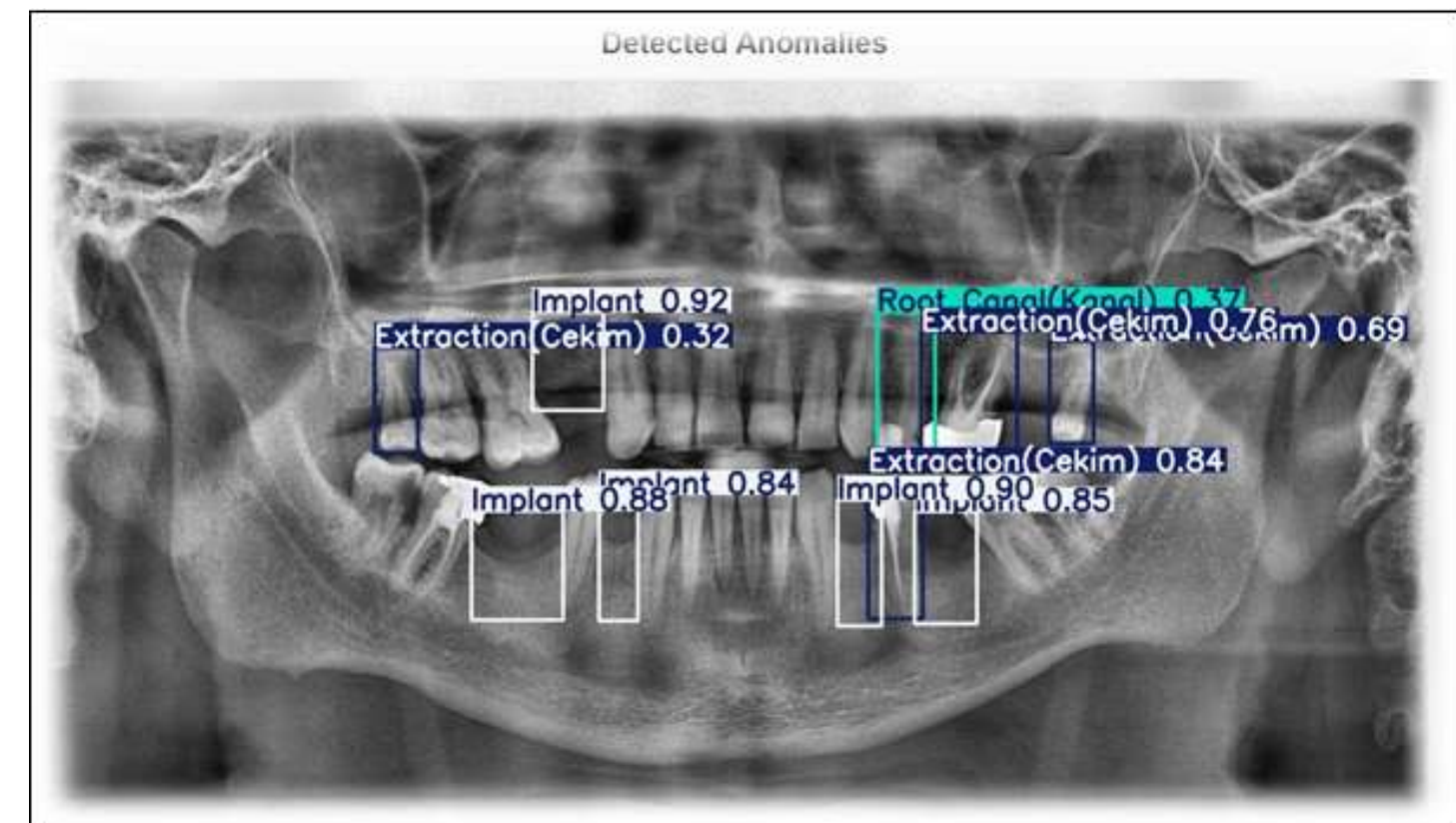
LOGIN →




Drag dental X-ray here
or click to browse files

Supported formats: JPG, JPEG, PNG

Choose File No file chosen



How about Ethical Perspective ?

The AI algorithm has the potential to perform even better than humans which needs:

- 1) Big data collection.
- 2) Quality of labeling

The system firstly will be considered as an assistant for the doctor.



EU Certifications already exists and some companies in market like Pearl which is raised \$69.25M and is backed by investors like the American Dental Association

Future Work...

In Summary, The system needs a Research which by estimation can be done in one year.

Requires a team of healthcare professionals and Data Scientists in order to deliver a market ready product.

AI can be enhanced to provide a health report using LLM explaining the issues in human language.



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https://github.com/apodwikat/Dental_Anomaly_Detection_GraduationProject